

NORRBERG-NORRHOG V, Sweden

WMO: 023370

Lat: 62.2632N

Lon: 15.6709E

Elev: 320

StdP: 97.54

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -24.9	(d) -30.7	(e) 0.2	(f) -27.9	(g) -27.4	(h) 0.3	(i) -24.9	(j) 4.9	(k) -0.5	(l) 4.3	(m) -2.0	(n) 0.1	(o) 320	(p) 0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	26.5	15.7	24.3	15.2	22.0	14.2	17.5	23.1	16.3	21.2	15.3	19.8	1.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	11.5	18.7	14.5	10.7	17.7	13.4	10.0	16.5	50.5	23.1	46.9	21.4	43.8	19.9	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.8	(b) 4.1	(c) 3.4	(e) -29.8	(f) 29.6	(g) 3.2	(h) 1.8	(i) -32.0	(j) 30.9	(k) -33.9	(l) 31.9	(m) -35.7	(n) 32.9	(o) -38.0	(p) 34.2		
(a) 4.8	(b) 4.1	(c) 3.4	(e) -30.2	(f) 19.3	(g) 2.8	(h) 0.5	(i) -32.2	(j) 19.7	(k) -33.8	(l) 20.0	(m) -35.4	(n) 20.3	(o) -37.5	(p) 20.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-9.4	-7.8	-3.2	2.6	7.3	11.7	14.6	12.9	7.9	2.2	-2.8	-7.5		
	DBStd	9.46	7.17	6.28	4.99	3.28	3.46	3.25	2.99	3.01	3.39	3.94	5.00	7.37		
	HDD10.0	3103	602	498	408	223	99	18	2	8	79	241	384	542		
	HDD18.3	5808	861	731	667	473	342	201	121	168	313	499	634	800		
	CDD10.0	344	0	0	0	1	16	68	144	99	16	0	0	0		
	CDD18.3	7	0	0	0	0	0	1	6	1	0	0	0	0		
	CDH23.3	276	0	0	0	0	12	88	134	42	0	0	0	0		
	CDH26.7	44	0	0	0	0	1	13	25	6	0	0	0	0		
(14) Wind	WSAvg	1.2	1.0	1.1	1.5	1.4	1.6	1.3	1.0	0.8	0.9	1.2	1.1	1.0		
(15) Precipitation	PrecAvg	610	39	27	27	35	45	67	88	82	54	56	48	42		
	PrecMax	880	71	50	63	112	108	123	204	138	143	140	123	67		
	PrecMin	491	6	3	5	7	15	29	29	46	14	18	17	5		
	PrecStd	92	18	13	13	25	25	24	42	29	30	30	27	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	6.0	10.9	19.0	24.9	28.3	29.2	27.2	20.6	13.7	9.4	6.0		
		MCWB	2.3	2.9	5.2	10.3	12.0	14.8	17.7	17.1	13.4	9.9	7.8	3.8		
	2%	DB	2.7	3.9	7.9	15.0	21.0	26.0	26.8	24.4	17.8	11.7	7.2	4.3		
		MCWB	1.1	1.8	3.4	7.9	11.5	14.7	16.6	16.1	12.2	8.9	5.8	2.7		
	5%	DB	1.2	2.3	6.0	12.7	17.9	23.2	24.6	22.0	15.8	10.0	5.1	3.2		
		MCWB	0.2	0.6	2.4	6.4	10.0	13.5	15.6	15.2	11.4	8.0	4.2	1.8		
	10%	DB	-0.1	0.6	4.1	10.3	15.1	20.0	22.3	19.8	13.9	8.6	3.5	1.7		
		MCWB	-0.8	-0.6	1.4	5.1	9.1	12.2	14.8	14.3	10.5	7.2	2.7	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.1	6.0	11.2	13.8	16.6	19.1	18.6	14.6	10.8	7.9	4.2		
		MCDB	4.4	5.0	10.3	16.2	21.5	24.1	26.5	23.3	18.2	12.6	9.4	5.2		
	2%	WB	1.4	2.1	3.7	8.7	12.6	15.3	17.8	17.5	13.2	9.5	5.9	3.1		
		MCDB	2.5	3.9	7.0	14.4	18.7	22.9	24.5	22.0	15.9	10.8	7.0	4.1		
	5%	WB	0.3	0.7	2.7	6.8	11.0	14.1	16.7	16.2	12.2	8.5	4.3	1.9		
		MCDB	0.9	1.9	5.5	11.5	16.7	21.7	21.9	20.2	14.8	9.8	5.0	3.0		
	10%	WB	-0.8	-0.5	1.6	5.4	9.7	12.9	15.6	15.1	11.2	7.1	2.9	0.6		
		MCDB	-0.2	0.5	3.8	9.7	14.5	18.7	20.3	18.5	13.3	8.6	3.6	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.7	11.4	11.7	12.4	13.2	12.7	12.2	10.8	8.5	6.5	7.7		
		MCDBR	7.2	8.5	12.5	17.7	18.4	20.2	17.9	16.9	13.9	10.7	7.5	7.2		
		MCWBR	6.4	6.7	8.5	10.5	9.4	9.0	8.3	8.8	8.5	7.8	6.5	6.2		
		MCDBR	7.0	8.0	11.6	16.0	15.5	17.0	15.4	13.5	11.6	9.8	7.0	7.1		
	5% WB	MCWBR	6.5	6.5	8.0	9.7	8.7	8.0	8.0	7.8	7.7	7.7	6.3	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.277	0.303	0.338	0.343	0.338	0.354	0.346	0.314	0.299	0.238	0.344		
	taud		2.111	2.264	2.338	2.400	2.451	2.493	2.460	2.477	2.555	2.477	2.188	2.723		
	Ebn at Noon		474	691	812	852	877	888	861	838	807	674	446	309		
	Edh at Noon		32	60	81	94	97	95	96	87	67	50	30	20		
(44) All-Sky Solar Radiation	RadAvg		0.22	0.86	2.34	3.89	4.68	5.36	4.88	3.73	2.36	1.05	0.30	0.12		
	RadStd		0.03	0.13	0.23	0.33	0.48	0.40	0.58	0.33	0.29	0.17	0.06	0.01		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page